WEEK 04 / EXPLAINABLE ARTIFICIAL INTELLIGENCE

Explainable AI and Model Interpretability

Global feature reliance and local explanations for a breast-tumour classifier

Technical report | Week 3 model refit | Reproducible interpretation

30	3	4 / 5
Features inspected globally	Local case studies	Local top-5 reference overlap

Educational demonstration only — not validated for diagnosis or patient-care decisions.

Executive summary

This report explains the predictions of the support vector classifier selected in Week 3 using two complementary techniques: permutation feature importance for global model reliance and Kernel SHAP for individual predictions. The model is refitted from scratch with the same training split and selected parameters rather than loaded from a saved model. Its held-out performance matches the earlier run: 99.12% accuracy and 98.80% malignant F1, with one false negative among 114 test cases. These scores describe this fixed split, not clinical effectiveness.

For global analysis, each feature was shuffled 30 times in the held-out test set, and the change in malignant F1 was recorded. 'Worst texture' had the largest observed mean F1 decrease, 0.0731. For local analysis, Kernel SHAP decomposed the model's signed decision score relative to a 40-record training reference. In the false-negative example, the reference score was -0.2960, the feature contributions summed to -0.8106, and the predicted score was -1.1066. A second training reference changed the attributions while preserving the prediction; four of the top five features overlapped. These explanations are descriptive model audits, not causal explanations of disease.

1. Objective, data and model under analysis

The aim is to make a previously trained predictive approach easier to inspect, not to claim that an explanation makes the model safe. The task is supervised binary classification on the Wisconsin Diagnostic Breast Cancer dataset: a model receives 30 numeric measurements derived from digitized fine-needle aspirate images of breast masses and predicts benign or malignant. The public scikit-learn copy has 569 records, including 357 benign and 212 malignant cases, with no missing feature values. Features summarize characteristics such as radius, texture, perimeter, area, smoothness and concavity. [1, 2]

As in Week 3, the original scikit-learn labels are reversed so that benign is 0 and malignant is the positive class, 1. A stratified 80/20 split with random seed 42 gives 455 training and 114 held-out test records. The classifier is a pipeline consisting of StandardScaler and an RBF support vector classifier with C=10, gamma=0.01 and balanced class weights. These are the hyperparameters selected by the Week 3 five-fold grid search; this Week 4 script does not repeat the search. Instead, it reconstructs and fits that same pipeline on the same training partition. The model has not been adjusted after examining the explanations. [3]

The SVC's decision_function returns a signed score: for these binary predictions a positive score is on the malignant side of the zero decision boundary and a negative score is on the benign side. This score is neither a calibrated probability nor a percentage risk. A false negative is a malignant case

classified as benign; this error motivates the local case study. The high internal test score is useful context for interpretation but is based on a small, historical dataset and one split.

Held-out measure	Recorded result	Context
Accuracy	99.12%	113 of 114 predictions correct
Malignant precision	100.00%	41 true positives / 41 predicted positives
Malignant recall	97.62%	41 of 42 malignant cases detected
Malignant F1	98.80%	Precision–recall harmonic mean
Confusion counts	TN 72, FP 0, FN 1, TP 41	Rows: actual; columns: predicted

2. Why two interpretability techniques?

A global explanation asks which inputs the fitted model relies on across a group of examples. A local explanation asks how input values contribute to a particular output relative to a reference. The questions are different: a feature that is influential for one case need not have the largest average effect across the dataset, and a global ranking cannot explain every individual decision. Using both perspectives enables a reviewer to move from a broad audit to the examination of a specific error without pretending that one chart describes the entire model.

Permutation feature importance is model-agnostic and assesses the change in a chosen evaluation metric after disrupting a feature. Kernel SHAP is also model-agnostic, but estimates additive contributions for individual model outputs. LIME is another possible local surrogate method; it was not implemented or compared in this experiment. The two implemented methods were selected because they address different levels of interpretation and can operate on the whole preprocessing-plus-SVC pipeline, even though the RBF SVM has no simple set of coefficients in the original feature space. [4, 5]

Both methods are post-hoc: they inspect a model that is already fitted rather than making its internal decision rule inherently transparent. The test cases were not used to fit the scaler, choose the Week 3 hyperparameters or train the classifier. They are used here for post-hoc global scoring and for selecting local examples, so they are no longer a fully blind resource for a future redesign. If an investigation leads to feature removal or model changes, those decisions must be developed on training/validation data and checked on a new independent test cohort.

3. Global technique: permutation feature importance

Permutation importance begins with the fitted model's baseline score on an evaluation dataset. For one feature, its values are randomly shuffled across rows while the remaining columns and labels stay in place. The fitted pipeline then predicts again, and the selected metric is recomputed. The importance for that shuffle is the baseline metric minus the shuffled metric. Repeating this operation produces an average drop and a standard deviation for each feature. A large positive drop indicates that this fitted model relied on that column for the selected metric on this particular dataset; a value near zero means that individual shuffling did not appreciably change that metric. [4]

In this experiment the evaluation set is the 114-record held-out test set, the scoring metric is malignant F1, and each of the 30 features is shuffled 30 times with seed 42. The whole pipeline receives the shuffled data in original units, then applies its already-fitted scaler before SVC prediction. The model and scaler are never retrained during permutation. This choice makes the

result relevant to the same positive class used for Week 3 model selection. A different metric, such as recall or ROC AUC, could produce a different ranking because it asks a different question.

F1 is calculated from hard predictions at the model's existing boundary, so small score shifts may have no effect unless cases cross that boundary. Repeated shuffling measures variation due to random permutations of these same records; the plotted standard deviation is not a confidence interval for the population and does not account for a new training set, a new hospital or a different model. A negative importance is possible if a shuffle happens to improve the score; it should prompt investigation rather than be treated as proof that the feature causes harm.

Feature	Mean F1 drop	Shuffle SD
worst texture	0.0731	0.0247
worst concavity	0.0467	0.0126
worst symmetry	0.0333	0.0102
texture error	0.0266	0.0089
mean concave points	0.0260	0.0161

The unshuffled malignant F1 was 0.9880. The largest average drop came from worst texture (0.0731 ± 0.0247), followed by worst concavity (0.0467) and worst symmetry (0.0333). For worst texture, the mean drop corresponds to approximately 7.31 F1 percentage points on this fixed test set. This does not mean that the feature accounts for that percentage of disease risk, nor that it is the most biologically important characteristic.

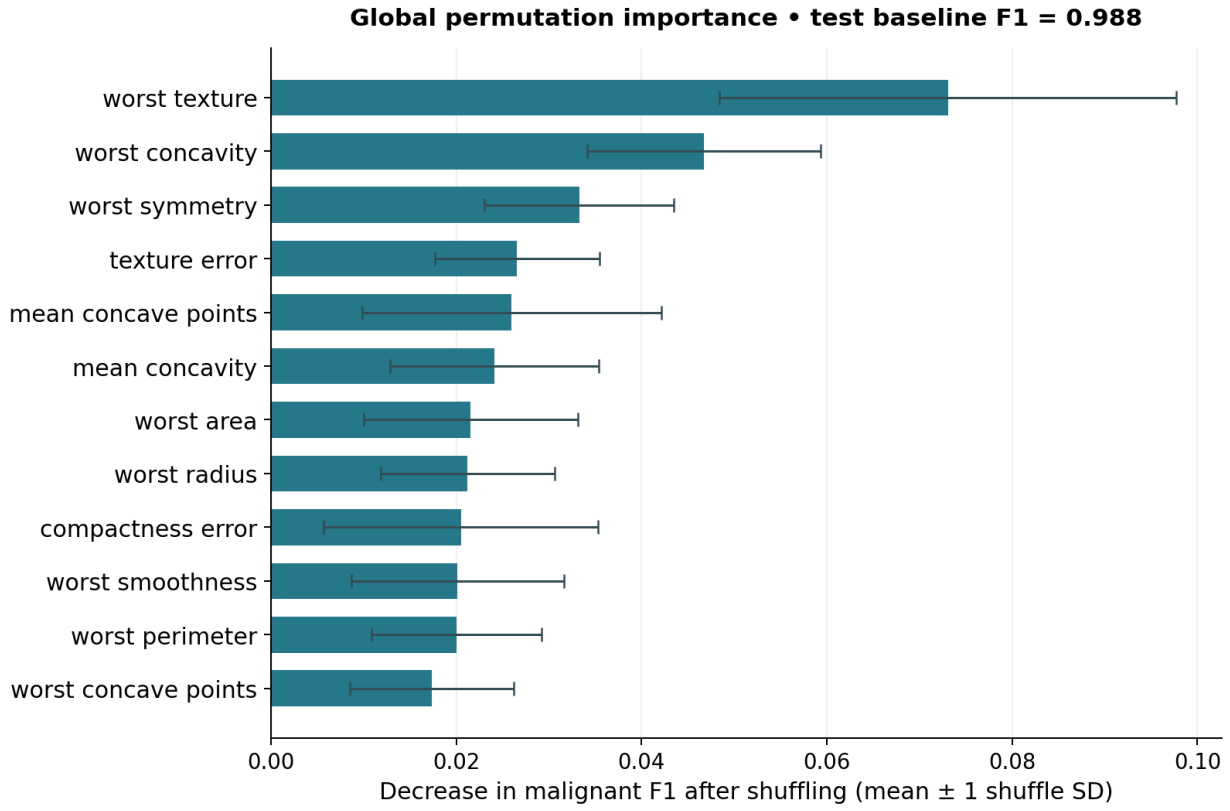


Figure 1. Global permutation importance for the 12 highest-ranked features. Bars are mean F1 drops over 30 shuffles; whiskers are ±1 shuffle standard deviation, not confidence intervals.

The full output also shows zero mean F1 change for smoothness error and concave points error in these particular shuffles. This does not prove that the features are irrelevant: their influence on the

continuous score may not move a test case across the decision boundary, and correlated features can substitute for one another. Conversely, shuffling a single feature breaks relationships with other measurements and can generate combinations that are implausible for a real cell image. The ranking is therefore an audit of model reliance under a specified perturbation, not an experiment that changes a real tumour. [6]

Correlation is not just a theoretical concern here. In the 455 training records, the Pearson correlation between worst radius and worst perimeter is 0.9937. A model may distribute signal across such related features, so shuffling one at a time can understate their joint importance. The code reports this descriptive correlation using training data only. Grouped or conditional permutation could be explored later, but neither is implemented in the submitted code.

4. Local technique: Kernel SHAP

SHAP is based on Shapley values from cooperative game theory. Conceptually, each feature receives an average contribution across possible orders in which features could be added to a coalition. For a specified value function, the contribution of feature j compares the model output with and without that feature in each coalition; the weighted average distributes credit among interacting features. Kernel SHAP approximates this process using a specially weighted local linear regression, enabling it to explain functions such as an SVC decision score without requiring access to the model's internal structure. [5, 7]

For one case, the additive form is f(x) = reference + sum of the 30 feature contributions. Here f(x) is the signed SVC decision score, and the identity link is used. The reference is the average model score over 40 real training records selected by stratified sampling: 25 benign and 15 malignant. The background is not the test case, a model probability or a target clinical population. When a feature is treated as absent in a coalition, Kernel SHAP replaces missing parts with values from the reference records and averages the model outputs. Holding some case features fixed while substituting others can create unrealistic combinations, especially when the features are correlated. [5, 8]

Enumerating all 2^30 coalitions would be impractical, so the script uses up to 1,024 coalition masks per case, with a fixed random seed of 123. It sets l1_reg=0.0 rather than using automatic top-ten feature selection, and records all 30 estimated attributions. A positive SHAP value raises the model score toward the malignant side relative to the reference; a negative value lowers it toward the benign side. The size is expressed in model-score units, not probability points, and the sign is not a statement about biological causation.

The script checks that reference plus all estimated contributions reconstructs each original model score within numerical tolerance. This is a useful implementation check, but Kernel SHAP's additive constraint can hold even when estimated feature values are unstable, unrealistic or inappropriate for a particular user. Additivity should not be mistaken for proof that the model's decision is medically correct. For visual clarity, the local charts show the eight largest absolute contributions and the net sum of the other 22; that net can hide offsetting positive and negative effects.

5. Local case study: false negative

The fixed test split contains one false negative. It is the record at zero-based dataset row 73 (test position 16), selected because its actual class is malignant while the model predicts benign. These row indices are dataset positions, not patient identifiers. Reference A has an average score of -0.2960; the 30 estimated contributions sum to -0.8106; together they reproduce the model's score of -1.1066. The negative score lies on the benign side of the decision boundary, despite the actual malignant label.

The largest absolute contribution for this case is worst texture, with recorded value 20.86 and estimated contribution -0.5482. Mean texture contributes -0.2542, while texture error contributes +0.2607 in the opposite direction. These are explanations of this model output relative to reference A: they do not establish that any measurement caused cancer or that altering a measurement would change the underlying diagnosis. The difference between the two texture-related effects also shows why a simple statement such as 'texture is always benign' would be incorrect.

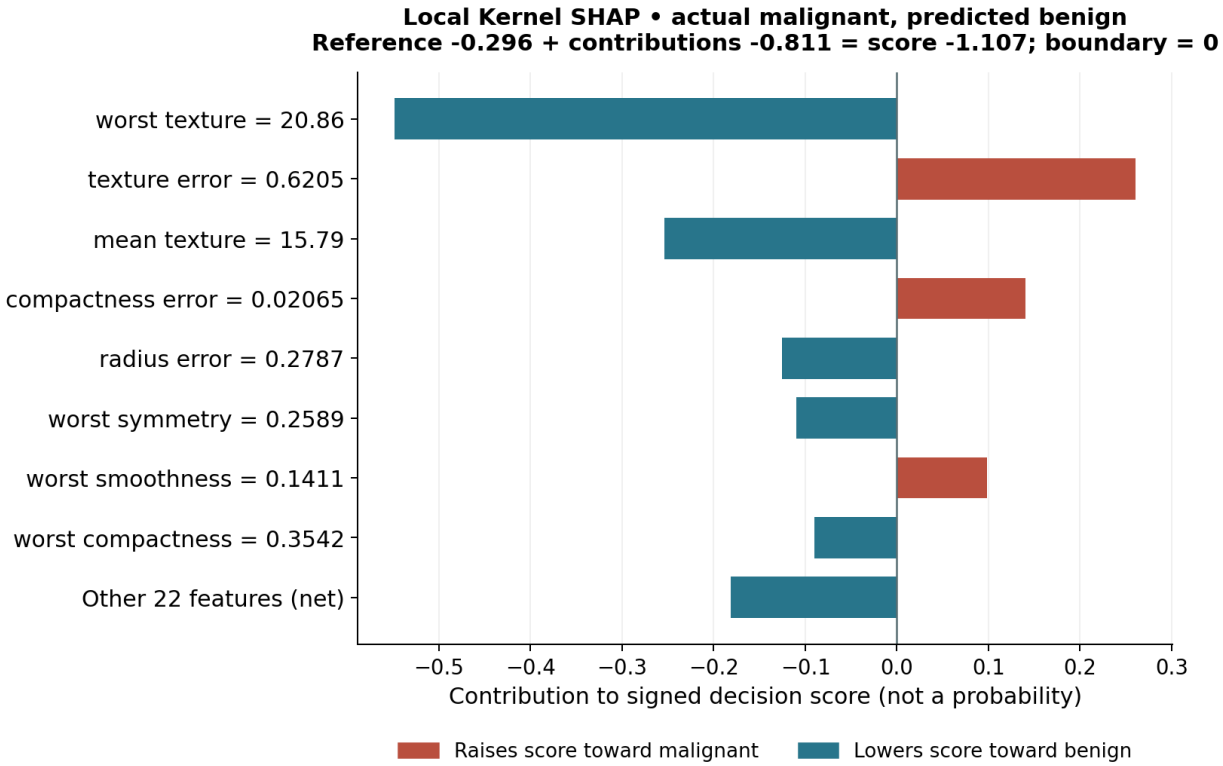


Figure 2. Kernel SHAP decomposition for the false-negative case under reference A. Positive bars raise the signed score toward malignant; negative bars lower it. The remaining 22 features are shown as a net sum.

A practical reviewer could use this chart to identify questions for an error investigation: whether the recorded measurements and label are correct, whether the feature-extraction process behaved unusually, and whether comparable false negatives recur in independent data. The chart cannot decide those questions by itself. In a genuine healthcare workflow, an error would require qualified review, independent validation and appropriate procedures rather than an automatic change to patient care.

6. Local comparison: correct predictions

The script also explains the first correctly predicted malignant case in the fixed test order, dataset row 250. Its reference is -0.2960; the summed contributions are +5.6204, producing a score of +5.3244, on the malignant side of zero. Its strongest estimated positive contributions include mean concave points (+0.5835), mean concavity (+0.5300) and worst area (+0.5204). The model combines many features, including the net contribution from the other 22 shown in Figure 3.

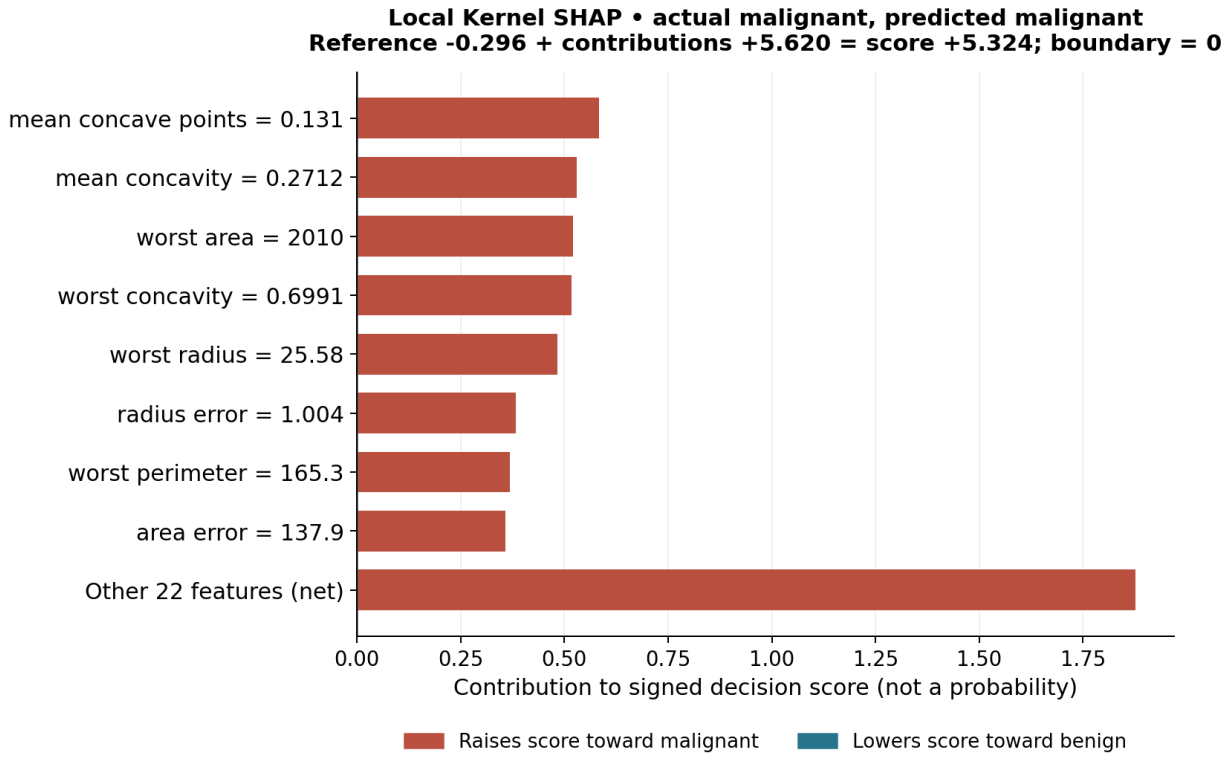


Figure 3. Kernel SHAP decomposition for the first true-positive test case under reference A. The score is not a probability; the last bar is the net of 22 features.

The first correctly predicted benign case, dataset row 120, has score -3.1700; its 30 values are included in the CSV even though a separate chart is not embedded. These three examples were selected by a deterministic rule, not sampled to represent all possible cases. The false negative was selected because it is an error, while the first true positive and true negative are illustrative comparisons. They cannot establish typical behaviour, fairness across groups or stability across future patients.

7. Background sensitivity experiment

An explanation depends on the reference population. To examine this dependence in a controlled example, a second set of 40 training records is selected with seed 7 instead of seed 42. Both references have the same class mix, 25 benign and 15 malignant, but only 3 records overlap. The fitted model, the false-negative input, the number of coalition masks and the coalition seed remain the same. The analysis is a two-reference sensitivity check, not a full uncertainty study.

False-negative quantity	Reference A	Reference B
Background mean score	-0.2960	-0.2463
Sum of 30 SHAP values	-0.8106	-0.8604
Reconstructed model score	-1.1066	-1.1066
Worst texture contribution	-0.5482	-0.4141

The model score remains -1.1066, but the baseline changes from -0.2960 to -0.2463, and the worst-texture attribution changes from -0.5482 to -0.4141. Across all 30 features, the mean absolute change is 0.0295 score units and the largest change is 0.1341. Four of the five largest absolute contributors are shared. The experiment demonstrates that a valid additive decomposition can change when the reference changes, even though the underlying prediction stays fixed.

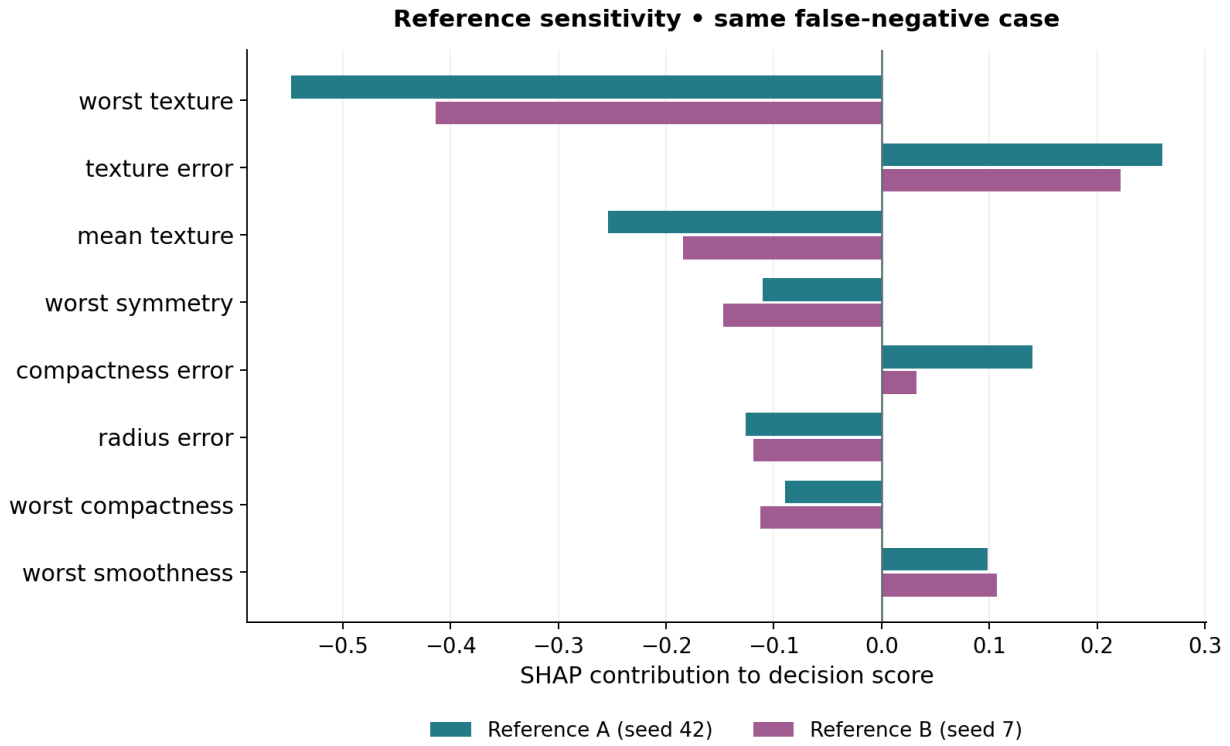


Figure 4. False-negative SHAP values under two different 40-record training references. The model, case and coalition seed are unchanged; this is not a confidence interval.

This comparison does not determine which reference is correct. The appropriate background depends on the question and intended use: a training-distribution reference, a clinically defined comparison cohort or another reference can yield different explanations. Larger or repeated reference samples, multiple coalition seeds and careful assessment of feature dependence would be needed to characterize sensitivity more fully. Any chosen reference should be documented and justified before results are shown to decision makers.

8. Interpreting global and local results together

Aspect	Permutation importance	Kernel SHAP
Question	Which column affects aggregate F1 when shuffled?	How is one signed score allocated relative to a reference?
Data and labels	Uses test features and labels for F1	Uses case features and a training reference; the score function needs no labels
Units	Change in malignant F1	Signed model-decision-score units
Main uncertainty	Shuffle randomness, finite test set and correlated columns	Background choice, sampled coalitions and dependent features

Worst texture appears both as the highest global permutation feature and as the strongest absolute local contribution for the false-negative case. That agreement is informative but not proof of causation or universal dominance. For the selected true positive, mean concave points has the largest estimated local effect, illustrating that the local explanation can differ from the global ranking. The two magnitudes cannot be compared numerically: an F1 change is a population-level metric difference, whereas a SHAP contribution is measured in the model's signed score units.

Neither method tells a user the model's complete reasoning in human terms. Permutation results depend on the score, examples and perturbation rule; SHAP results depend on the value function, reference distribution and approximation. Correlated inputs can share or exchange credit, and the

hypothetical masked inputs may be outside the observed data distribution. A method can be precisely reproduced and still answer a question that is poorly aligned with the intended decision. The correct interpretation is therefore conditional: this is what the fitted model appears to rely on under the stated procedures and data, not what medically causes a diagnosis. [6, 8]

9. Trust, accountability and real-world application

In an appropriately governed real-world setting, global importance can support model documentation and data-quality checks. For example, an unexpectedly dominant acquisition-related variable might lead a team to inspect collection practices or test for distribution shift. Local explanations can help prioritize the review of errors, reveal conflicts between model outputs and domain expectations, and communicate which measured features moved a score relative to a stated reference. The correct next step is investigation by qualified people, not automatic acceptance of the explanation.

Accountability requires more than producing attractive charts. A deployment team should record the model version, training-data provenance, evaluation set, feature definitions, label mapping, chosen metric, decision threshold, explanation method, background sample and software version. Logs should protect sensitive information through access controls and data-minimization practices. A reviewer should be able to trace an explanation back to the prediction, contest it, report a failure and obtain human oversight. Changes to the model or reference distribution should trigger renewed validation and documentation.

Explanations do not establish fairness, and this dataset does not provide the demographic and institutional information needed for a meaningful subgroup audit in the model features. If such information were collected lawfully and with appropriate protections, separate analyses could examine group-specific sensitivity, false-positive rates, uncertainty and missingness, followed by stakeholder review. A feature that is not itself a protected attribute can still act as a proxy. Nor does a plausible explanation guarantee security, reliability or good clinical outcomes; independent evaluation and governance remain necessary.

Finally, the displayed inputs are measurements of a mass, not levers that an individual can safely modify. A negative SHAP value is not medical reassurance, and a positive value is not a diagnosis. Decisions involving people require validated clinical pathways, expert interpretation, safeguards and compliance with applicable rules. This educational model and its explanations must not be used to diagnose or treat anyone.

10. Reproducibility, limitations and future work

The accompanying Python script loads the bundled dataset, recreates the original split and refits the selected Week 3 pipeline. It computes the baseline metrics, training-only correlation, 30-repeat permutation importance, three individual Kernel SHAP explanations and a second-reference check for the false negative. A numerical additivity assertion checks each local decomposition. Two executions in the recorded environment produced identical numerical outputs. Reproducibility under one fixed seed and software environment is not the same as robustness across new datasets or settings.

```
python -m pip install numpy scipy scikit-learn matplotlib shap
python week4_xai_model.py --output-dir results
```

The program writes metrics.json, permutation_importance.csv, local_shap_values.csv and five PNG figures. The first CSV contains all 30 features and every shuffle result; the second contains 30 contributions for each of the three cases plus the alternate-reference false-negative explanation. metrics.json records the split, selected model, class encoding, seeds, case selection, reference sizes, feature rankings, reconstruction checks and package versions. The run used Python 3.13.16, NumPy 2.3.5, SciPy 1.17.1, scikit-learn 1.6.1, SHAP 0.54.1 and Matplotlib 3.10.9. Other versions may give slightly different outputs.

Important limitations remain: a single dataset, a small held-out set, three chosen case studies, two reference samples and a finite number of sampled coalitions. Shuffle standard deviations are not population confidence intervals, and the reference comparison does not isolate every source of estimator uncertainty. The high correlations between image-derived measurements complicate independent masking. Explanations were not tested with clinicians, and no group fairness analysis or prospective external validation was performed.

Future work could repeat global analyses across resampled datasets and independent cohorts; compare F1, recall and ROC AUC importances; and investigate grouped or dependence-aware methods. For local SHAP, repeated coalition seeds and multiple justified reference cohorts could quantify variability. A clinician-facing study could evaluate whether users understand the explanation and identify model failures without becoming overconfident. These are proposals, not experiments already completed. Any model redesign should be evaluated on a fresh, previously unseen external set.

11. Conclusion

The Week 3 RBF SVC was reconstructed and inspected with complementary global and local XAI techniques. Permuting worst texture produced the largest observed reduction in test F1, while Kernel SHAP exposed the combination of positive and negative score contributions for an individual false negative. The background experiment showed that local attributions can change even when the model's output does not. Together, the results support a careful audit trail and focused error review, but they do not provide causal explanations, guarantee fairness or justify clinical use.

References

[1] Wolberg, W., Mangasarian, O., Street, N. and Street, W. Breast Cancer Wisconsin (Diagnostic) [dataset]. UCI Machine Learning Repository, 1995. https://doi.org/10.24432/C5DW2B. Dataset licence: CC BY 4.0.

[2] scikit-learn developers. load_breast_cancer: Breast cancer Wisconsin diagnostic dataset. https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

[3] scikit-learn developers. SVC: Support vector classification. https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html

[4] scikit-learn developers. Permutation feature importance. https://scikit-learn.org/stable/modules/permutation_importance.html

[5] SHAP developers. shap.KernelExplainer: model-agnostic Kernel SHAP. https://shap.readthedocs.io/en/latest/generated/shap.KernelExplainer.html

[6] scikit-learn developers. Permutation importance with multicollinear or correlated features. https://scikit-learn.org/stable/auto_examples/inspection/plot_permutation_importance_multicollinear.html

[7] Lundberg, S. M. and Lee, S.-I. A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems, 2017. https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[8] Aas, K., Jullum, M. and Løland, A. Explaining individual predictions when features are dependent. Artificial Intelligence 298, 103502, 2021. https://doi.org/10.1016/j.artint.2021.103502